

Continuity (9.3)

Name _____

Circuit

Begin with the problem in the top left. Answer the question. To advance in the circuit, find your answer and write “2” in the blank. Solve that problem, find your answer and write “3” in the blank and so on until you have solved every problem and returned to the beginning.

ANSWER: 3

_____ At what x -value is $f(x) = \frac{x^2-1}{x-1}$ discontinuous?

ANSWER: -3

_____ The function $f(x) = \lfloor x \rfloor$ is discontinuous at every _____.

ANSWER: 1

_____ At what x -value is $f(x) = \frac{1}{x-7}$ discontinuous?

ANSWER: yes

_____ A piecewise function has left value 3 and right value 5 at $x = 2$. Is it continuous there? (yes/no)

ANSWER: 0 # _____ The nonremovable discontinuity of $f(x) = \frac{x-2}{x^2-5x+6}$ is at $x = ?$	ANSWER: 2 # _____ Is $f(x) = x^2$ continuous for all real numbers? (yes/no)
ANSWER: $f(a)$ # _____ At what x-value is $f(x) = \frac{5}{x}$ discontinuous?	ANSWER: 7 # _____ How many discontinuities does $f(x) = \frac{x}{x^2-4}$ have?
ANSWER: no # _____ At what x-value does $f(x) = \frac{x^2-9}{x+3}$ have a hole?	ANSWER: integer # _____ If f is continuous at $x = a$, then $\lim_{x \rightarrow a} f(x)$ must equal what?